

Business Problem Statement

PowerGrid Utilities Corporation is one of the largest electricity transmission and distribution providers in the region, serving millions of residential, commercial, industrial, and government customers through an extensive network of substations, transformers, transmission lines, distribution assets, and smart grid infrastructure. Over the past decade, increasing urbanization, industrial growth, renewable energy integration, and climate-related disruptions have significantly increased the complexity of managing grid reliability and service continuity.

Executive leadership has become increasingly concerned about the rising frequency of power outages, equipment failures, voltage instability events, and service interruptions affecting customers across multiple service territories. While utility operators continuously monitor network performance through supervisory control systems and smart grid technologies, several recent large-scale outages have resulted in significant financial losses, regulatory scrutiny, customer dissatisfaction, and reputational damage. Internal investigations indicate that many of these incidents were preceded by operational warning signals that existed within maintenance systems, asset monitoring platforms, weather forecasting tools, and grid performance databases but were not acted upon in time.

Multiple departments have raised concerns regarding grid resilience and operational reliability. Maintenance teams report increasing volumes of aging infrastructure requiring inspection and replacement. Network operations teams have identified abnormal load patterns and capacity utilization concerns across critical substations. Asset management teams have observed growing transformer degradation risks. Customer service teams have reported increasing complaint volumes during peak demand periods. Environmental and regulatory teams have highlighted growing weather-related risks associated with extreme heat, storms, flooding, and other climate events. Leadership believes that valuable insights may be hidden within operational telemetry, equipment health metrics, maintenance records, weather information, customer demand patterns, outage histories, and grid asset performance data.

Currently, operational decisions are largely reactive, with interventions often occurring after service degradation or equipment failures have already impacted customers. Leadership wants to transition toward a predictive grid management strategy capable of identifying high-risk assets and operating conditions before major service interruptions occur. The analytics team has been provided with historical operational data collected from smart grid systems, maintenance platforms, outage management systems, weather monitoring services, and customer operations databases. The dataset contains information related to equipment health, power demand, weather conditions, maintenance activities, operational performance, outage histories, and several legacy administrative attributes inherited from previous utility systems. Management has intentionally not prescribed a specific analytical approach and expects the analytics team to investigate the available information, identify meaningful patterns, uncover factors associated with grid failures, and recommend data-driven solutions that can improve reliability, reduce outages, and strengthen infrastructure resilience.